QUIZ_2

How much voltage is dropped across R2 and R3 in Figure 4-1 if R1 = 4.7 kΩ, VR1 = 10 V, R2 = 4.7 kΩ and R3 = 4.7 kΩ? Question 1 options: VR2 = 10 V, VR3 = 4.7 V VR2 = 4.7 V, VR3 = 10 V VR2 = 10 V, VR3 = 10 V VR2 = 14.7 V, VR3 = 14.7 V. . . . 2
When the resistance and voltage are known, what is the formula for finding the current (amperes)? Question 2 options: I = R/V I = V/R I = V x R I = V x R. 3
When a 24-volt battery is connected to a series circuit with three resistors of 1 ohm, 3 ohms, and 2 ohms, how many amperes of current will flow through the circuit? Question 3 options: 1 3 4 2. 4
A 4.7 kΩ resistor has 3 mA of current passing through it. The resistor dissipates Question 4 options: 42.3 W 35.2 mW 42.3 mW 14.1 mW. 5
The sum of the voltage drops in a series circuit equals the: Question 5 options: Source voltage Total resistance Current in circuit Resistance in branch. 6
7
milli (m) means which of these? Question 7 options: 1 100 103 10-3. 8
The formula for total resistance (RT) in a series circuit is which of these? Question 8 options: RT = R1 - R2 RT = R1 X R2 X R3 RT = RT = R1 + R2 + R3. 9
kilo (k) means which of these? Question 9 options: 103 10-3 100 1. 10
A series circuit consists of three resistors with values of 1.1 kΩ, 4.4 kΩ, and 2.5 kΩ across a 30 V source. The current through the 2.5 kΩ resistor is Question 10 options: 2.24 µA 3.75 mA 3.75 A 2.24 mA. 11

How much voltage is dropped across R2 and R3 in Figure 4-1 if R1 = 4.7 kΩ, VR1 = 10 V, R2 = 4.7 kΩ and R3 = 4.7 kΩ?
Question 1 options:

VR2 = 10 V, VR3 = 4.7 V

VR2 = 4.7 V, VR3 = 10 V

VR2 = 10 V, VR3 = 10 V

VR2 = 14.7 V, VR3 = 14.7 V

Since all three resistors are equal and the same series current flows through each, the voltage drop across R_2 and R_3 equals that across R_1:

$$I = \frac{V_{R1}}{R1} = \frac{10\ \text{V}}{4.7\ \text{k}\Omega} \approx 2.13\ \text{mA},$$

$$V_{R2} = I\,R2 = 2.13\ \text{mA} \times 4.7\ \text{k}\Omega = 10\ \text{V},$$

$$V_{R3} = I\,R3 = 2.13\ \text{mA} \times 4.7\ \text{k}\Omega = 10\ \text{V}.$$

Answer: VR_2 = 10 V, VR_3 = 10 V.

When the resistance and voltage are known, what is the formula for finding the current (amperes)?

Question 2 options:

I = R/V

I = V/R

I = V x R

I = V x R

The correct formula is Ohm's Law solved for current:

$$I = \frac{V}{R}$$

So the answer is **I = V/R**.

When a 24-volt battery is connected to a series circuit with three resistors of 1 ohm, 3 ohms, and 2 ohms, how many amperes of current will flow through the circuit?

Question 3 options:

1

3

4

2

The total series resistance is

$$R_{\text{total}} = 1 + 3 + 2 = 6\ \Omega.$$

By Ohm's Law,

$$I = \frac{V}{R_{\text{total}}} = \frac{24\ \text{V}}{6\ \Omega} = 4\ \text{A}.$$

Answer: 4 amperes.

A 4.7 kΩ resistor has 3 mA of current passing through it. The resistor dissipates
Question 4 options:

42.3 W

35.2 mW

42.3 mW

14.1 mW

The power dissipated is

$$P = I^2R = (0.003\ \text{A})^2 \times 4700\ \Omega = 9 \times 10^{-6} \times 4700 = 0.0423\ \text{W} = 42.3\ \text{mW}.$$

Answer: 42.3 mW.

The sum of the voltage drops in a series circuit equals the:

Question 5 options:

Source voltage

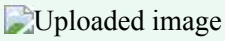
Total resistance

Current in circuit

Resistance in branch

The sum of the voltage drops in a series circuit equals the **source voltage**.





The three 10 kΩ resistors are all in series, so

$$R_{\text{total}} = R_1 + R_2 + R_3 = 10\,\text{k}\Omega + 10\,\text{k}\Omega + 10\,\text{k}\Omega = 30\,\text{k}\Omega.$$

milli (m) means which of these?
Question 7 options:

1

100

103

10-3

The prefix “milli” (m) denotes a factor of 10^{-3}.

The formula for total resistance (RT) in a series circuit is which of these?

Question 8 options:

RT = R1 - R2

RT = R1 X R2 X R3

RT =

RT = R1 + R2 + R3

The total resistance in a series circuit is the sum of the individual resistances:

$$\boxed{R_T = R_1 + R_2 + R_3}$$

kilo (k) means which of these?

Question 9 options:

103

10-3

100

1

The prefix “kilo” (k) denotes a factor of 10^3.

A series circuit consists of three resistors with values of 1.1 kΩ, 4.4 kΩ, and 2.5 kΩ across a 30 V source. The current through the 2.5 kΩ resistor is

Question 10 options:

2.24 μA

3.75 mA

3.75 A

2.24 mA

Total series resistance:

$$R_{\text{tot}} = 1.1\,\text{k}\Omega + 4.4\,\text{k}\Omega + 2.5\,\text{k}\Omega = 8.0\,\text{k}\Omega.$$

Current through any series resistor:

$$I = \frac{V}{R_{\text{tot}}} = \frac{30\,\text{V}}{8.0 \times 10^{3}\,\Omega} = 3.75 \times 10^{-3}\,\text{A} = 3.75\,\text{mA}.$$

Answer: 3.75 mA.